

BELLABEAT

Smart Device Usage Analysis

Data Analytics Case Study

Prepared by	Darshana Narjari
GitHub Repository	bellabeat-data-analytics-case-study
Tools	Excel, Power Query, SQL (PostgreSQL), Power BI
Selected Bellabeat Product	Bellabeat App
Project Start Date	17-09-2026
Project Completion Date	19-09-2026

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1. Business Understanding

Scenario

I am a junior data analyst on Bellabeat's marketing analytics team.

My role is to analyze smart-device usage data and present high-level recommendations that can help inform Bellabeat's marketing strategy.

Business background

Bellabeat is a high-tech company that develops health-focused smart products designed specifically for women. Its products and services help users track and understand areas of their health and wellness, such as physical activity, sleep, stress, hydration, and mindfulness.

Bellabeat operates in the global smart-device and wellness technology market. The company aims to use technology and health-related data to help women better understand their habits and make healthier decisions.

This analysis is being conducted to identify trends in how consumers use smart devices. The insights from the analysis will help Bellabeat understand consumer behavior and develop high-level recommendations that can support and improve its marketing strategy

Characters and stakeholders

Primary stakeholder: Urška Sršen - Bellabeat cofounder and Chief Creative Officer.

Secondary stakeholder: Sando Mur - Bellabeat cofounder and key executive team member.

Business team: Bellabeat Marketing Analytics Team.

Bellabeat products

The case study introduces these products: Bellabeat app, Leaf, Time, Spring, and Bellabeat Membership.

Selected product for this project: Bellabeat App

Reason for selecting this product: I selected the Bellabeat App because it is directly connected to users' health and wellness data, including activity and sleep-related information. The available smart-device dataset contains similar types of information, allowing me to analyze consumer behavior and explore how the insights could be applied to Bellabeat's app and marketing strategy

Business task statement

Bellabeat wants to better understand how consumers use smart devices for tracking their health and wellness. I will analyze publicly available Fitbit smart-device usage data to identify trends and patterns in areas such as physical activity, sleep, and daily habits. The objective is to use these insights to understand how similar trends

could apply to Bellabeat customers and develop high-level recommendations that can help Bellabeat's growth & marketing strategy.

2. Data Preparation

Data source register

Data source name: FitBit Fitness Tracker Data

Original link: Kaggle- [Kaggle dataset source](#)

License: CC0: Public Domain

Date accessed: September 17, 2026

Why this source was selected: This dataset was selected because it contains real-world smart-device usage data, including physical activity, steps, calories, activity intensity, sleep, heart rate, and weight. These variables are relevant to understanding consumer health and wellness behavior and can be used to identify trends that may provide insights for Bellabeat's marketing strategy.

Data credibility worksheet (ROCCC)

Reliable: The dataset contains personal fitness-tracker data submitted by 30 eligible Fitbit users who consented to share their data. However, the sample is relatively small and may not represent the broader population of wearable-device users.

Original: The dataset consists of fitness-tracker records collected from Fitbit users, including activity, heart rate, and sleep information. It is not data that I created or simulated for this analysis.

Comprehensive: The dataset contains multiple categories of health and activity information, including daily, hourly, and minute-level data for physical activity, calories, steps, intensity, heart rate, and sleep. However, not every user provided every type of information, so some datasets contain fewer users than others.

Current: The dataset is not current because the data was collected between March 12, 2016 and May 12, 2016. Therefore, the findings may not fully represent current consumer behavior or today's wearable-device market.

Cited: The dataset is publicly available through Kaggle and includes an acknowledgement of the original researchers and source. I will cite the dataset source and original research information in the references section of my project.

Guiding questions from the case study

Are there issues with bias or credibility in this data? — Yes. The dataset has some limitations related to bias and credibility. The sample is relatively small and consists of Fitbit users who agreed to share their data. Therefore, the participants may not represent all wearable-device users. In addition, the dataset represents Fitbit users rather than Bellabeat customers, so the findings cannot be assumed to directly represent Bellabeat customers.

How are you addressing licensing, privacy, security, and accessibility? -- The dataset is publicly available through Kaggle and is listed under the CC0 Public Domain license. I will use the data only for this educational portfolio project and will provide appropriate source attribution.

How does the data help you answer the business question? -- The dataset contains information about physical activity, steps, calories, activity intensity, and sleep. These variables can help me identify patterns in how people use smart devices to monitor their health and daily habits. The identified patterns can then be considered in the context of Bellabeat's customers and used to develop insights that may support Bellabeat's marketing strategy.

Dataset inventory

File Name	Purpose	Rows	Columns	Used? (Y/N)
dailyActivity_merged.csv	Daily activity, steps, distance, activity minutes and calories	940	15	Y
dailyCalories_merged.csv	Daily calories burned	940	3	Y
dailyIntensities_merged.csv	Daily activity intensity and intensity-related distance	940	10	Y
dailySteps_merged.csv	Daily step totals	940	3	Y
sleepDay_merged.csv	Sleep duration and time in bed	413	5	Y
weightLogInfo_merged.csv	Weight, BMI and body-fat information	67	8	N

Data dictionary template

Column Name	Data Type	Meaning	Example	Notes
Id	Integer	Unique identifier for a Fitbit user	1503960366	Anonymous user identifier
ActivityDate	Date	Date on which activity was recorded	4/12/2016	Used to analyze activity by date
TotalSteps	Integer	Total number of steps recorded during the day	13162	Important activity metric
TotalDistance	Decimal	Total distance traveled during the day	8.5	Distance recorded by the tracker
Calories	Integer	Total calories burned during the day	1985	Used to study energy expenditure
VeryActiveMinutes	Integer	Minutes spent in very active activity	25	Activity intensity measure
FairlyActiveMinutes	Integer	Minutes spent in fairly active activity	13	Activity intensity measure
LightlyActiveMinutes	Integer	Minutes spent in light activity	328	Activity intensity measure
SedentaryMinutes	Integer	Minutes spent sedentary	728	Used to study inactive behavior
SleepDay	Date/Time	Date associated with the sleep record	4/12/2016	Sleep dataset
TotalMinutesAsleep	Integer	Total minutes the user was asleep	327	Used for sleep analysis
TotalTimeInBed	Integer	Total minutes spent in bed	346	Can be compared with actual sleep time
BMI	Decimal	Body Mass Index recorded for the user	22.65	Available only for users with weight records

Data limitations worksheet

Known limitation 1: The dataset was collected in 2016 and may not represent current smart-device usage behavior.

Known limitation 2: The dataset contains a relatively small number of Fitbit users and does not represent the entire population of wearable-device users.

Impact on conclusions: The findings should be interpreted as directional insights rather than conclusions that represent all consumers or Bellabeat customers. The results can help identify possible trends and marketing

opportunities, but additional and more current data would be needed to validate the findings for Bellabeat customers

3. Data Cleaning & Transformation

Tools used

Excel: Used for initial data inspection and basic data checks.

PostgreSQL: Used for querying, filtering, aggregating, and analyzing the cleaned data.

Power Query: Used for data cleaning, transformation, and combining datasets.

Power BI: Used to create the final dashboard and visualizations.

Data cleaning checklist

- ☒ Checked row and column counts
- ☒ Checked duplicate records
- ☒ Checked missing values
- ☒ Checked inconsistent text values
- ☒ Checked date formats
- ☒ Checked numeric data types
- ☒ Checked invalid or impossible values
- ☒ Standardized column names
- ☒ Verified joins/relationships if combining datasets
- ☒ Saved cleaned version separately

Cleaning log

Issue Found	Affected File	Action Taken	Reason	Verified?
Data type inconsistencies	dailyActivity_merged, dailyCalories_merged, dailyIntensities_merged, dailySteps_merged, sleepDay_merged, dailyActivity_merged (2)	Converted text to date, decimal and numeric types.	To ensure accurate analysis	Yes
Duplicate records	sleepDay_merged	Removed duplicate records	Prevent duplicate sleep observations	Yes
Separate data periods	dailyActivity_merged	Appended March–April and April–May data	Create one activity dataset	Yes

Transformation log

Transformation	Tool Used	Before	After	Reason
Convert date columns	Power Query	Text/incorrect type	Date	Enable date-based analysis
Remove duplicate records	Power Query	Duplicate rows	Unique records	Prevent double-counting
Append activity datasets	Power Query	Two periods	One combined dataset	Create continuous activity data
Remove unnecessary columns	Power Query	Extra fields	Relevant fields only	Simplify analysis

Integrity verification

Dataset	Before	After
Daily Activity	1,397	1,397
Daily Calories	940	940
Daily Intensities	940	940
Daily Steps	940	940
Sleep Day	413	410

Excel / Power Query documentation

Worksheet names used: dailyActivity_merged, dailyCalories_merged, dailyIntensities_merged, dailySteps_merged, sleepDay_merged

Queries created: dailyActivity_Combined, dailyCalories_Cleaned, dailyIntensities_Cleaned, dailySteps_Cleaned, sleepDay_Cleaned

Important transformations: Changed data types, checked missing values and duplicates, removed unnecessary columns, and appended the two daily activity periods.

SQL import documentation

Database name: Bellabeat

Tables created: daily_activity, sleep_day, daily_calories, daily_intensities, daily_steps

Import method: CSV files imported into PostgreSQL using pgAdmin.

Import validation performed: Verified row counts, column names, data types, missing values, and duplicate records.

4. Exploratory Data Analysis

Analysis Approach :- The analysis was conducted using Excel and PostgreSQL to identify patterns in daily activity, sedentary behavior, and sleep. The analysis focused on overall activity levels, differences between weekdays and weekends, sleep patterns, user-level behavior, and the relationship between physical activity and sleep.

Overall Activity :- The dataset contained 35 unique users and 1397 daily activity records. The average daily step count was 7280.90 steps, while average daily calorie expenditure was 1363.19 calories. Users spent an average of 992.54 minutes per day being sedentary. Activity intensity analysis showed that users spent more time in **lightly activity**.

Weekday vs Weekend Behavior :- Daily activity was compared between weekdays and weekends to identify differences in user behavior. The analysis compared average steps, calories, and sedentary minutes across the two day types.

Average daily steps were **7317.02 on weekdays** compared with **7188.26 on weekends**. Sedentary time was **995.83 minutes on weekdays** and **984.09 minutes on weekends**.

This suggests that activity patterns vary between weekdays and weekends.

Sleep Patterns :- Sleep behavior was analyzed using total minutes asleep and total time in bed. The analysis also calculated sleep efficiency by comparing actual sleep time with time spent in bed.

Users slept an average of **419.17 minutes per day** and spent approximately **458.48 minutes in bed**.

Average sleep efficiency was **91.65%**.

Activity and Sleep Relationship :- To investigate whether physical activity was associated with sleep, daily activity records were matched with sleep records using the user's ID and date. A correlation analysis was performed between total daily steps and total minutes asleep.

The correlation between daily steps and minutes asleep was **-0.19**. This indicates a negative relationship, meaning higher step counts tended to be associated with slightly lower sleep duration in this dataset.

However, correlation does not establish causation, and the result should be interpreted as an association within this dataset.

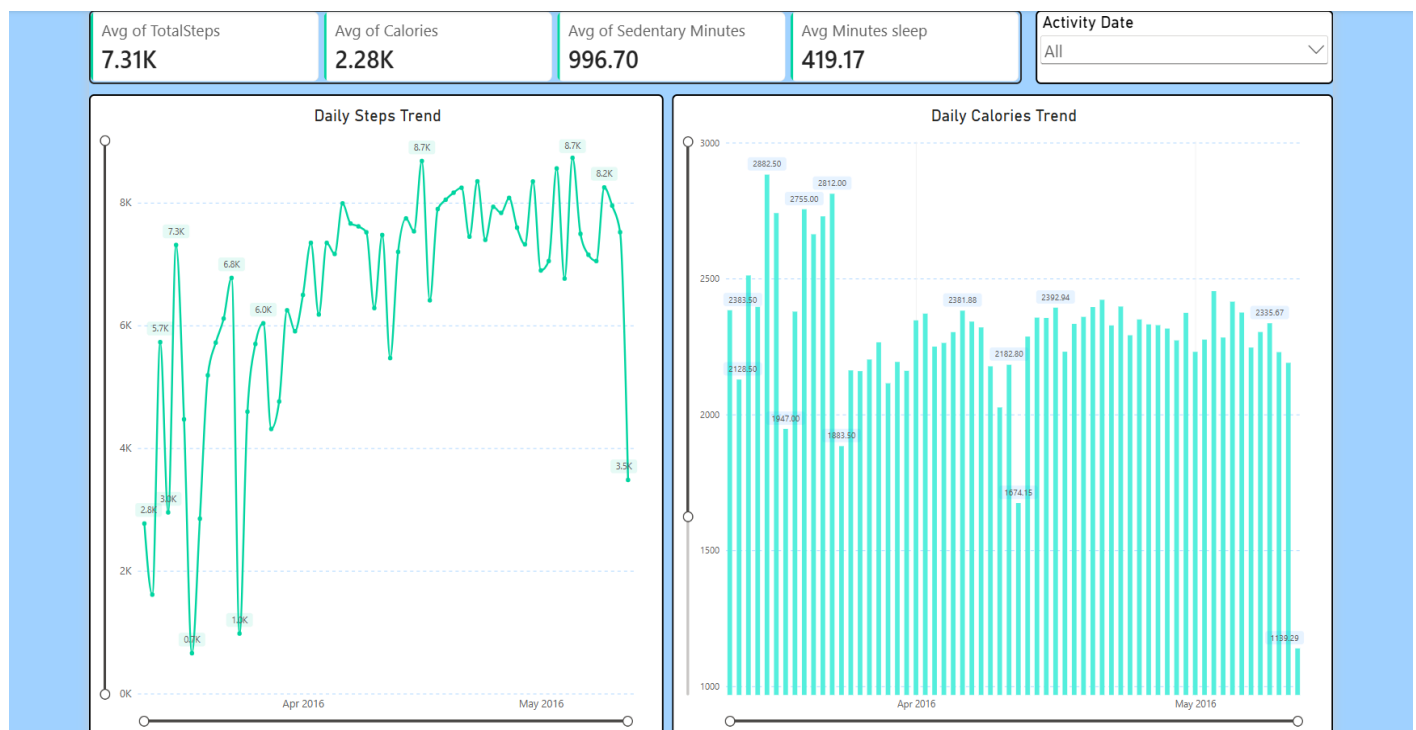
5. Data Visualization & Insights

Power BI Dashboard

An interactive Power BI dashboard was developed to communicate the key findings from the analysis. The dashboard is organized into three pages: Overview, Activity Behavior, and Sleep & Wellness. The visuals summarize activity levels, sedentary behavior, calorie expenditure, sleep patterns, and the relationship between physical activity and sleep.

Overview Dashboard

The Overview dashboard provides a high-level summary of Fitbit user activity and wellness behavior. Key metrics include average daily steps, average daily calories burned, average sedentary minutes, and average minutes asleep. Daily trend charts were also used to observe changes in steps and calorie expenditure over the observation period.



Activity Behavior Dashboard

The Activity Behavior dashboard examines differences in physical activity and sedentary behavior. It compares weekday and weekend activity, summarizes time spent at different activity intensities, and categorizes daily activity levels using total steps.



Sleep & Wellness Dashboard

The Sleep & Wellness dashboard examines sleep duration, time spent in bed, sleep efficiency, and the relationship between daily steps and sleep duration. These visuals help provide a broader view of wellness behavior beyond physical activity alone.



Key Findings

Finding 1 - Daily activity

The dataset showed an average daily step count of approximately **7,311 steps** across the full activity dataset.

Finding 2 - Sedentary behavior

Users recorded substantial sedentary time. The weekday/weekend comparison also showed that **average weekday steps were higher while average weekday sedentary minutes were also higher.**

This suggests that higher daily movement and sedentary behavior can occur on the same day.

Finding 3 - Activity intensity

Light activity and sedentary behavior represented a substantial portion of recorded daily behavior, while higher-intensity activity accounted for considerably fewer minutes.

Finding 4 - Sleep

Average recorded sleep was approximately **419 minutes**, or about **7 hours per day**.

Finding 5 - Activity and sleep relationship

The matched activity and sleep records produced a correlation of approximately: **-0.19**

This indicates a **weak negative relationship** between daily steps and sleep duration in this dataset. As one increases other decreases. It does not establish causation.

6. Recommendations & Conclusion

- **Encourage Everyday Movement**

Bellabeat could use its app to encourage consistent daily movement through personalized step goals, walking challenges, activity streaks, and movement reminders.

Marketing opportunity:

Campaigns could focus on the idea that small, consistent movements throughout the day contribute to overall wellness rather than focusing only on intensive workouts.

- **Reduce Prolonged Sedentary Time**

The analysis showed that weekday users had both **higher average step counts and higher average sedentary minutes** than weekends. This indicates that being more active during a day does not necessarily mean spending less time sedentary.

Recommendation:

Bellabeat could introduce features that encourage users to take regular movement breaks, such as reminders to walk, stretch, or change position after extended periods of inactivity.

- **Connect Activity and Sleep Insights**

The analysis examined both physical activity and sleep behavior. The relationship between daily steps and sleep duration was weak, with a correlation of **-0.19**, indicating that the two measures did not have a strong linear relationship in this dataset.

Recommendation:

Bellabeat could combine activity and sleep information within the app to provide users with a broader daily wellness summary.

For example, the app could display: **Daily Activity → Sleep → Wellness Summary**

Conclusion

The analysis identified patterns in physical activity, sedentary behavior, and sleep that could provide directional insights for Bellabeat's marketing strategy. The recommendations focus on encouraging everyday movement, increasing awareness of sedentary behavior, and integrating activity and sleep insights.

Future Analysis

Future analysis using Bellabeat's own customer data, a larger sample, and more recent observations could validate these findings and support more targeted marketing strategies.

7. PORTFOLIO & GITHUB PACKAGING

Portfolio Case Study Structure

The completed project will be organized as a portfolio case study containing:

- Final Bellabeat Case Study Report (PDF)
- Power BI Dashboard (.pbix)
- SQL Analysis File (.sql)
- Cleaned Data Files
- Power BI Dashboard Screenshots
- README Documentation
- References

GitHub Repository

Repository Name: bellabeat-data-analytics-case-study

The GitHub repository will contain the project report, SQL analysis, Power BI dashboard, cleaned data, and dashboard visuals in an organized structure.

README

The README will provide a brief overview of the project, including:

- Project objective
- Business task
- Data source
- Tools used
- Analysis process
- Key findings
- Recommendations
- Project files

Portfolio Purpose

This project demonstrates practical skills in Excel, Power Query, SQL, and Power BI, including data cleaning, analysis, visualization, and business insight generation.

8. REFERENCES

1. Fitbit Fitness Tracker Data: [Kaggle dataset source](#) Kaggle. Public dataset containing daily activity, sleep, calories, steps, intensity, and other Fitbit user data.
2. Bellabeat Case Study: Google Data Analytics Professional Certificate. *Bellabeat Case Study: How Can a Wellness Technology Company Play It Smart?* Used as the business case and project brief.
3. PostgreSQL Documentation: PostgreSQL Global Development Group. *PostgreSQL Documentation*. Used as a reference for SQL queries and analytical functions.
4. Microsoft Power BI Documentation: Used as a reference for dashboard creation, data modeling, visualizations, and analysis.
5. Microsoft Excel / Power Query Documentation: Microsoft. *Microsoft Support - Excel and Power Query Documentation*. Used as a reference for data cleaning, transformation, and preparation.

9. APPENDICES

Appendix A - Dataset inventory

File Name	Purpose	Rows	Columns	Notes
dailyActivity_merged.csv	Daily activity, steps, distance, active minutes, sedentary minutes and calories	1,397 raw / 1,373 unique user-days	15	Combined two source periods; overlapping user-date records reviewed
dailyCalories_merged.csv	Daily calorie expenditure	940	3	Used for calorie analysis
dailyIntensities_merged.csv	Daily activity intensity and intensity minutes	940	10	Used to analyze activity intensity
dailySteps_merged.csv	Daily step totals	940	3	Used to support step analysis
sleepDay_merged.csv	Daily sleep duration and time in bed	410 cleaned	5	3 exact duplicate records removed from 413 original rows

Appendix B - Data dictionary

Column	Type	Definition	Example	Notes
Id	Integer	Unique identifier assigned to each Fitbit user	1503960366	Used to identify users across datasets
ActivityDate	Date	Date on which daily activity was recorded	2016-04-12	Used for daily activity analysis
TotalSteps	Integer	Total number of steps recorded during the day	10,245	Used as a key physical activity metric
Calories	Integer	Total calories burned during the day	1,980	Used for calorie analysis
SedentaryMinutes	Integer	Minutes spent in sedentary activity	720	Used to examine sedentary behavior
TotalMinutesAsleep	Integer	Total minutes recorded as asleep	420	Used for sleep-duration analysis
TotalTimeInBed	Integer	Total minutes spent in bed	450	Used to calculate sleep efficiency